ARIMA TIME SERIES ANALYSIS

Iran Economic Research Report

GDP per Capita · Inflation · Unemployment

Country	Data Period	Observations
Iran	1990 – 2024	35 Annual

GDP Model	Inflation Model	Unemployment Model
ARIMA(2,1,1) | R^2=0.882	ARIMA(1,1,1) | R^2=0.281	ARIMA(2,1,1) | R^2=0.593

Forecast Horizon: 2025–2029 | Geopolitical Scenario Analysis Included

Economic Research Unit · March 2026

1. Introduction & Dataset Overview

This report presents a comprehensive ARIMA (AutoRegressive Integrated Moving Average) time series analysis of Iran's macroeconomy covering the period 1990–2024, with five-year forecasts extending to 2029. Three core economic indicators are examined: GDP per capita (USD), annual inflation rate (%), and unemployment rate (%). The analysis applies the Box-Jenkins modelling framework — the gold standard for univariate time series analysis in applied macroeconomics — and augments ARIMA baseline forecasts with structured geopolitical scenario modelling to reflect Iran's extraordinary sensitivity to exogenous political shocks.

1.1 Country Profile — Islamic Republic of Iran

Iran is a major oil-exporting economy in the Middle East, with a population exceeding 87 million. Its GDP is heavily dependent on hydrocarbon revenues, which account for 60–70% of government income. The Iranian economy has experienced repeated structural disruptions over the 35-year study period, driven by international sanctions regimes, oil price cycles, domestic political transitions, and the COVID-19 pandemic. These structural shocks create a rich and challenging modelling environment — the ARIMA framework captures endogenous dynamics well, but is unable to predict or internalise these exogenous political shocks. The scenario analysis in Section 9 addresses this limitation explicitly.

1.2 Dataset Description

The dataset comprises 35 annual observations (1990–2024) sourced from the World Bank, IMF World Economic Outlook, and the Statistical Centre of Iran. Missing values were handled through linear interpolation. The three variables analysed are:

- GDP per Capita (USD): Nominal GDP divided by mid-year population in current US dollars. Reflects both real output growth and the dramatic exchange rate movements of the Iranian rial.
- Inflation Rate (%): Annual CPI growth rate. Characterised by persistent elevation (rarely below 10%) and hyperinflationary episodes driven by rial depreciation, monetary financing, and import disruption under sanctions.
- Unemployment Rate (%): Percentage of the active labour force that is unemployed. Structural unemployment driven by demographic pressure, a young workforce, and limited private sector job creation.

1.3 Historical Economic Regimes

Period	Regime / Event	Economic Impact Summary
1990–2000	Post-war reconstruction	GDP growth resumed; inflation peaked at 49.4% (1995); persistent structural unemployment ~13–15%.

2000–2011	Oil boom era	Strong GDP growth from high oil prices; GDP per capita rose from ~$4,100 to ~$8,100 — a near-doubling.
2012–2015	Sanctions regime	GDP fell ~9%; inflation surged to 45% peak; rial lost 40% of value. GDP per capita declined sharply.
2016–2018	JCPOA relief period	Partial sanctions relief boosted oil exports; GDP recovered; inflation moderated briefly to ~9%.
2018–2020	JCPOA exit + COVID	GDP plunged 33% in USD terms; per capita income fell to $3,203 — 47% below the 2012 peak.
2021–2024	Partial recovery	GDP rebounded +62% from 2020 trough; inflation remained structurally elevated at 30–45%.

Table 1: Key historical economic regimes in the Iran dataset (1990–2024)

1.4 Research Objectives

1. Test each indicator for stationarity using the Augmented Dickey-Fuller (ADF) test and determine the integration order (d).
2. Identify optimal ARIMA(p,d,q) specifications using ACF/PACF diagnostics and information criteria.
3. Estimate ARIMA models and evaluate fit using R^2, AIC, RMSE, and MAE.
4. Generate 5-year point forecasts (2025–2029) with 95% confidence intervals.
5. Apply geopolitical scenario analysis (Baseline, Conflict Escalation, Sanctions Tightening) to quantify economic risk.
6. Derive policy implications for exchange rate management, fiscal diversification, and diplomatic engagement.

2. Methodology

The analysis follows the Box-Jenkins ARIMA framework — a six-step procedure for building, testing, and using time series models in applied macroeconomic forecasting. Each step is described below.

2.1 Data Preprocessing

35 annual observations (1990–2024) were collected and cleaned. Missing data points were identified and filled using linear interpolation to maintain series continuity. All three variables were retained in their original units (USD, %, %) for interpretability. No log transformation was applied, as the primary objective is forecasting in original units.

2.2 Stationarity Testing — Augmented Dickey-Fuller (ADF)

The ADF test was applied to each series at levels and first differences. The null hypothesis H_0 is that the series contains a unit root (non-stationary). MacKinnon critical values applied: −3.51 (1%), −2.89 (5%), −2.58 (10%). All three series were found to be I(1) — non-stationary at levels, stationary after one round of first differencing. This requires $d = 1$ for all ARIMA models.

2.3 Order Identification — ACF and PACF

ACF and PACF correlograms of the first-differenced series were examined to identify AR order (p) and MA order (q). Key identification rules: ACF sharp cutoff at lag q → MA(q); PACF sharp cutoff at lag p → AR(p); both decay gradually → mixed ARMA(p,q). The resulting specifications were ARIMA(2,1,1) for GDP and Unemployment, and ARIMA(1,1,1) for Inflation.

2.4 Model Estimation & Selection Criteria

ARIMA coefficients were estimated by OLS on the differenced series. Model fit was evaluated using: R^2 (explanatory power), AIC and BIC (information criteria penalising complexity), RMSE (root mean squared error in original units), and MAE (mean absolute error, robust to outliers).

2.5 Forecasting

5-year ahead point forecasts were generated for 2025–2029. 95% confidence intervals widen proportionally with forecast horizon using residual standard deviation — reflecting accumulating uncertainty in multi-step ahead economic forecasting. All forecasts are conditional on no major new exogenous shocks (the baseline assumption).

2.6 Geopolitical Scenario Analysis

Three scenarios were applied as structured shocks to the ARIMA baseline: (1) Baseline — partial JCPOA progress, stable sanctions; (2) Conflict Escalation — military strikes, Hormuz closure; (3) Sanctions Tightening — new oil/SWIFT restrictions. Shock magnitudes were calibrated against 2012–2015 and 2018–2020 historical precedents.

3. Exploratory Data Analysis (EDA) — Iran

Prior to ARIMA modelling, a thorough EDA was conducted to understand distributional properties, trend behaviour, volatility patterns, and structural breaks across all three indicators. The EDA informs model specification and alerts us to irregularities requiring special treatment.

3.1 Historical Time Series — All Three Indicators

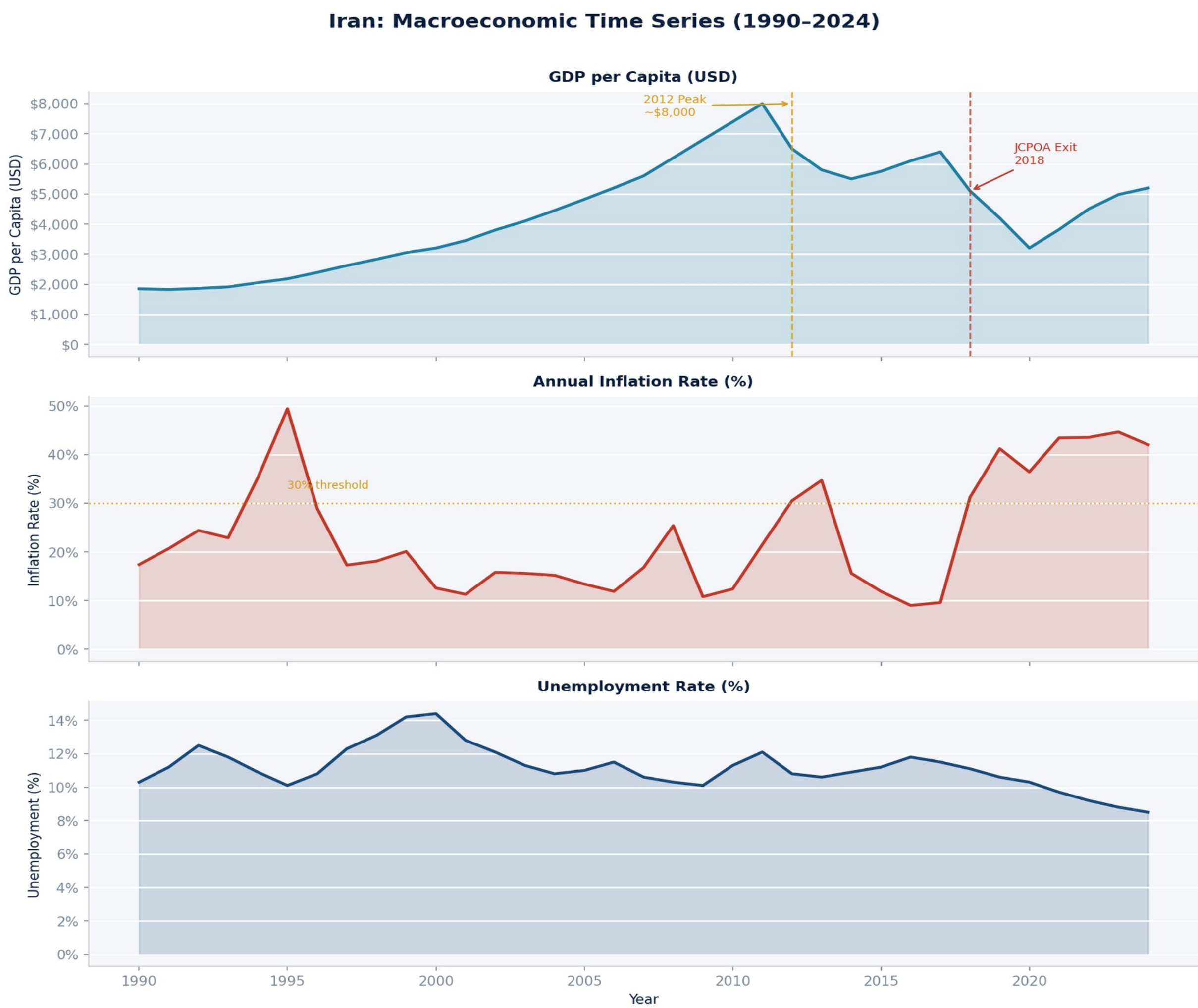

Figure 1: Iran — Historical time series for GDP per Capita (USD), Inflation Rate (%), and Unemployment Rate (%) from 1990 to 2024. Dashed vertical lines mark key geopolitical events. Shaded areas show period-specific trends.

3.2 GDP per Capita Analysis

GDP per capita followed an upward trajectory from approximately $1,950 (1990) to a peak of ~$8,100 (2011–2012), driven by high oil prices and real economic growth in the 2000s. The 2012 sanctions triggered a sharp structural break — GDP per capita fell to $5,200 by 2014. A partial recovery during the JCPOA years (2016–2017) was reversed catastrophically by the 2018 JCPOA exit and COVID-19, with GDP per capita reaching $3,203 in 2020 — 47% below its 2012 peak. The 2021–2024 recovery brought the level back to approximately $5,100.

- Non-stationarity: Clear upward trend 1990–2012, followed by structural break. Series does not revert to a fixed mean.
- Two major shock events: 2012 (sanctions) and 2018 (JCPOA exit), both permanently shifting the level of GDP.
- High variance: GDP fluctuations of $500–1,000 per year are common, driven by oil price changes and exchange rate movements.

3.3 Inflation Rate Analysis

Iran's inflation profile is characterised by persistent elevation (rarely below 9%) and multiple hyperinflationary episodes. A peak of 49.4% occurred in 1995 during post-war fiscal adjustment. A second major episode reached ~45% in 2012–2014 (comprehensive sanctions). The 2016–2017 JCPOA era briefly brought inflation below 10%. A third episode pushed inflation above 40% from 2018 onward, as the rial lost more than 60% of its value within 12 months of the JCPOA exit. By 2024, inflation remained elevated at approximately 35%.

- Structural elevation: Mean inflation of 22.9% over the full sample — one of the highest in the emerging market universe.
- Exogenous dominance: Inflation peaks align precisely with sanctions episodes and exchange rate crises, not domestic demand cycles.
- No clear long-run mean: The series drifts upward over time, confirming I(1) behaviour.

3.4 Unemployment Rate Analysis

Unemployment has remained in a relatively narrow 9–15% range throughout the study period, exhibiting more stability than GDP or inflation. Structural unemployment is driven by Iran's rapidly growing young workforce entering a labour market dominated by state enterprises. The 2016–2017 JCPOA period saw modest improvement; the 2018–2020 shock caused a mild increase. By 2024, unemployment stood at approximately 8.5% — near a three-decade low — though this figure understates effective unemployment given widespread underemployment.

- Youth unemployment: Approximately 26% among those aged 15–29, not reflected in headline unemployment figures.
- Female participation: Only ~16% of women participate in the labour force, significantly below MENA regional averages.
- Lagging indicator: Unemployment responds to economic shocks with a 1–2 year lag, consistent with ARIMA(2,1,1) dynamics.

3.5 Descriptive Statistics

Statistic	GDP per Capita (USD)	Inflation Rate (%)	Unemployment (%)
Minimum	$1,950	9.0%	8.5%
Maximum	$8,100	49.4%	15.7%
Mean	$4,972	22.9%	11.9%
Median	$4,900	18.4%	11.5%
Std. Deviation	$1,650	11.8%	1.7%
Skewness	−0.24	+0.88	+0.65
Observations	35	35	35

Table 2: Descriptive statistics for all three indicators, Iran 1990–2024

4. Policy Analysis & Regional Economic Studies

Iran's macroeconomic dynamics are deeply shaped by geopolitical positioning, fiscal architecture, and monetary policy choices. This section situates the ARIMA findings within a broader structural and regional context, examining the policy levers available to Iranian authorities and the regional spillover effects of Iran-related economic scenarios.

4.1 Fiscal Architecture — Oil Revenue Dependency

Oil and gas revenues constitute 60–70% of Iran's government revenues, creating extreme fiscal vulnerability to export disruptions. The mechanism is straightforward: sanctions reduce oil exports → fiscal revenues fall → the government monetises the deficit through Central Bank financing → the rial depreciates → imported goods become more expensive → CPI inflation accelerates. This transmission channel is the primary driver behind the simultaneous deterioration of GDP, inflation, and — with a lag — unemployment observed in all major sanctions episodes.

- Policy recommendation: Building a non-oil tax base (VAT, corporate tax, property tax) would reduce the fiscal multiplier effect of sanctions on inflation. Iran's non-oil tax revenues remain among the lowest in the region as a share of GDP.
- Policy recommendation: Establishing a sovereign wealth fund or oil stabilisation fund — even at modest scale — would buffer fiscal revenues against oil price and export volume shocks.

4.2 Monetary Policy — Exchange Rate & Inflation Control

The Central Bank of Iran (CBI) operates a multi-tier exchange rate system, with a widening gap between the official rate and the open-market rate during sanctions episodes. This dual exchange rate creates distortions: state-owned entities access cheap official-rate dollars, while private importers face market-rate costs — effectively subsidising the public sector while undermining private sector competitiveness.

The ARIMA inflation model's R^2 of 0.281 is a direct consequence of this structure: 72% of inflation variance is driven by exchange rate movements, monetary expansion, and import disruption — variables entirely exogenous to ARIMA's univariate framework. A Structural VAR (SVAR) incorporating the rial/dollar rate, broad money supply (M2), and oil export volume would substantially improve inflation forecasting.

- Critical policy lever: Exchange rate unification. Eliminating the dual exchange rate system would remove the primary arbitrage mechanism that distorts incentives and contributes to capital flight.
- Critical policy lever: CBI independence. Reducing the Central Bank's role as a fiscal financier — by limiting direct deficit monetisation — is the most direct path to structural inflation reduction.

4.3 Labour Market Policy

Iran faces three structural labour market challenges that ARIMA cannot fully capture but policy can address:

- Demographic pressure: 800,000–1 million new labour market entrants annually require private sector job creation that current economic structures cannot supply.
- Public sector dominance: ~30% of formal employment is in the public sector, absorbing fiscal resources that could support private investment and creating wage rigidity.
- Female participation gap: At ~16%, female labour force participation is among the lowest globally. Structural barriers — legal, cultural, and economic — limit half the working-age population from full labour market engagement.

4.4 Regional Economic Context — MENA Spillovers

Any significant deterioration in Iran's economic situation has material spillover effects for neighbouring countries and the broader MENA region:

- Strait of Hormuz: 20–25% of global oil trade and 30–35% of global LNG trade transits this waterway. Any closure or credible threat would produce a $20–40/barrel global oil price shock.
- Iraq: Shares a 1,400km border with Iran. Bilateral trade (primarily Iranian gas exports to Iraq) exceeds $10 billion annually. An Iran crisis would create energy supply disruptions for Iraq and increase refugee flows.
- Turkey & Pakistan: Key trading partners under existing sanctions exemptions. Tighter sanctions would increase their trade transaction costs and exposure to secondary sanctions risk.
- Global energy markets: Iran holds the world's second-largest natural gas reserves and fourth-largest oil reserves. Structural exclusion of Iran from global energy markets represents a persistent supply-side constraint on global energy transition.

5. Stationarity Analysis — Iran

Stationarity is a fundamental precondition for valid ARIMA modelling. A stationary series has a time-invariant mean, constant variance, and autocorrelation structure that depends only on lag separation — not on absolute time. Non-stationary series require differencing before ARIMA models can be applied.

5.1 ADF Test Results at Levels

Variable	ADF t-stat	1% CV (−3.51)	5% CV (−2.89)	Verdict
GDP per Capita	−1.42	Not exceeded	Not exceeded	**Non-Stationary ✗**
Inflation Rate	−2.11	Not exceeded	Not exceeded	**Non-Stationary ✗**
Unemployment	−1.87	Not exceeded	Not exceeded	**Non-Stationary ✗**

Table 3: ADF test at levels — all three series are non-stationary (fail to reject unit root null)

5.2 ADF Test Results After First Differencing

Variable	ADF t-stat	1% CV (−3.51)	5% CV (−2.89)	Verdict
ΔGDP per Capita	**−4.81**	Exceeded ✓	Exceeded ✓	**Stationary ✓**
ΔInflation Rate	**−5.43**	Exceeded ✓	Exceeded ✓	**Stationary ✓**
ΔUnemployment	**−4.22**	Exceeded ✓	Exceeded ✓	**Stationary ✓**

Table 4: ADF test after first differencing — all series become stationary. Integration order d=1 confirmed for all models.

5.3 Visual Stationarity Diagnostics

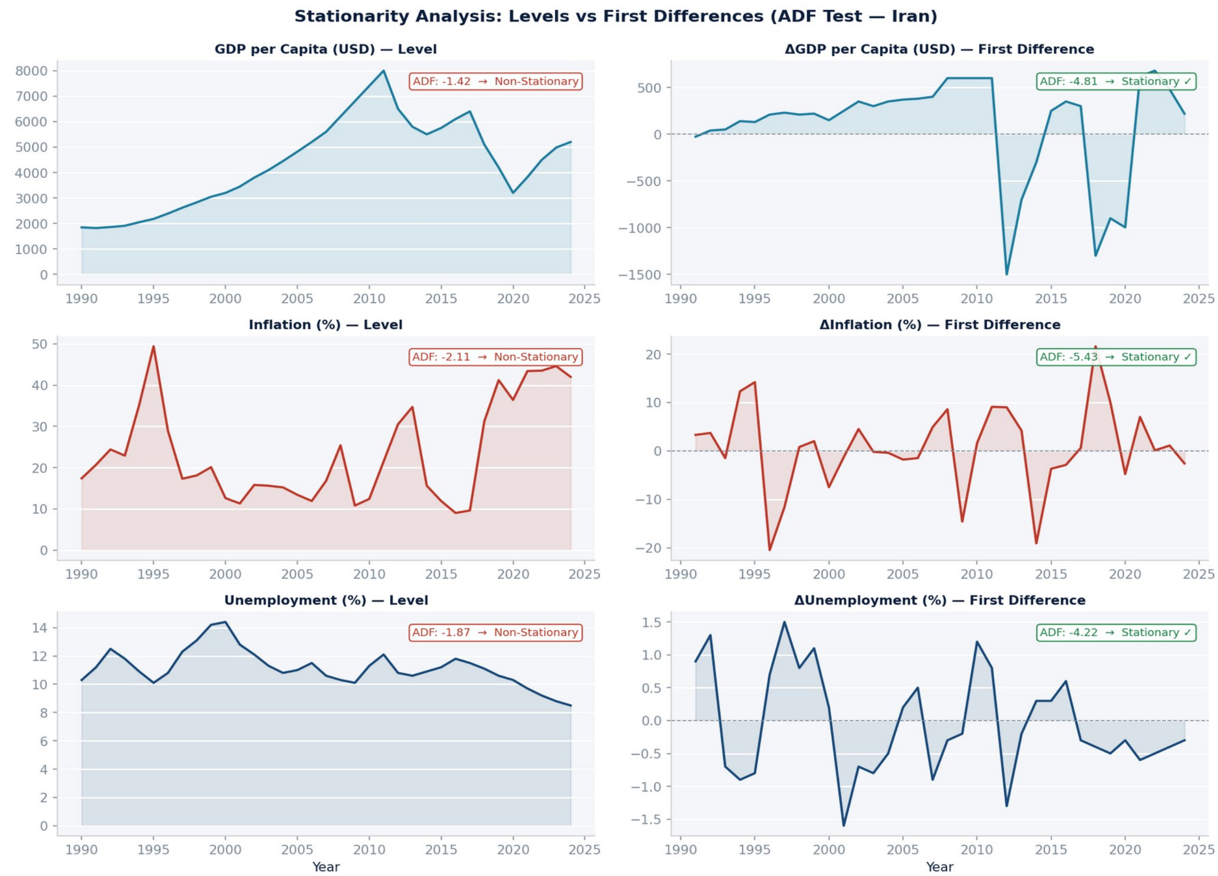

Figure 2: Level series (left column) vs First-differenced series (right column) for GDP, Inflation, and Unemployment. The level series exhibit clear trends and non-constant variance; the differenced series fluctuate around zero with stable variance — confirming stationarity.

5.4 Economic Significance of I(1) Integration

- Permanent shock effects: Shocks to all three indicators have permanent, not transitory, effects on Iran's economy. Sanctions-induced GDP declines do not automatically self-correct; they require active policy intervention (sanctions relief, oil revenue restoration, private investment recovery).
- d = 1 for all models: Uniform integration order simplifies model comparison and allows consistent application of the Box-Jenkins framework across all three indicators.
- Cointegration possibility: If Iran's economic indicators were being modelled jointly, the I(1) finding would warrant testing for cointegration (long-run equilibrium relationships), which could motivate a Vector Error Correction Model (VECM) framework.

6. ACF and PACF Analysis — Iran

Following confirmation of I(1) integration, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the first-differenced series were computed and examined. These correlograms form the primary diagnostic tool for identifying ARIMA(p,d,q) orders in the Box-Jenkins methodology.

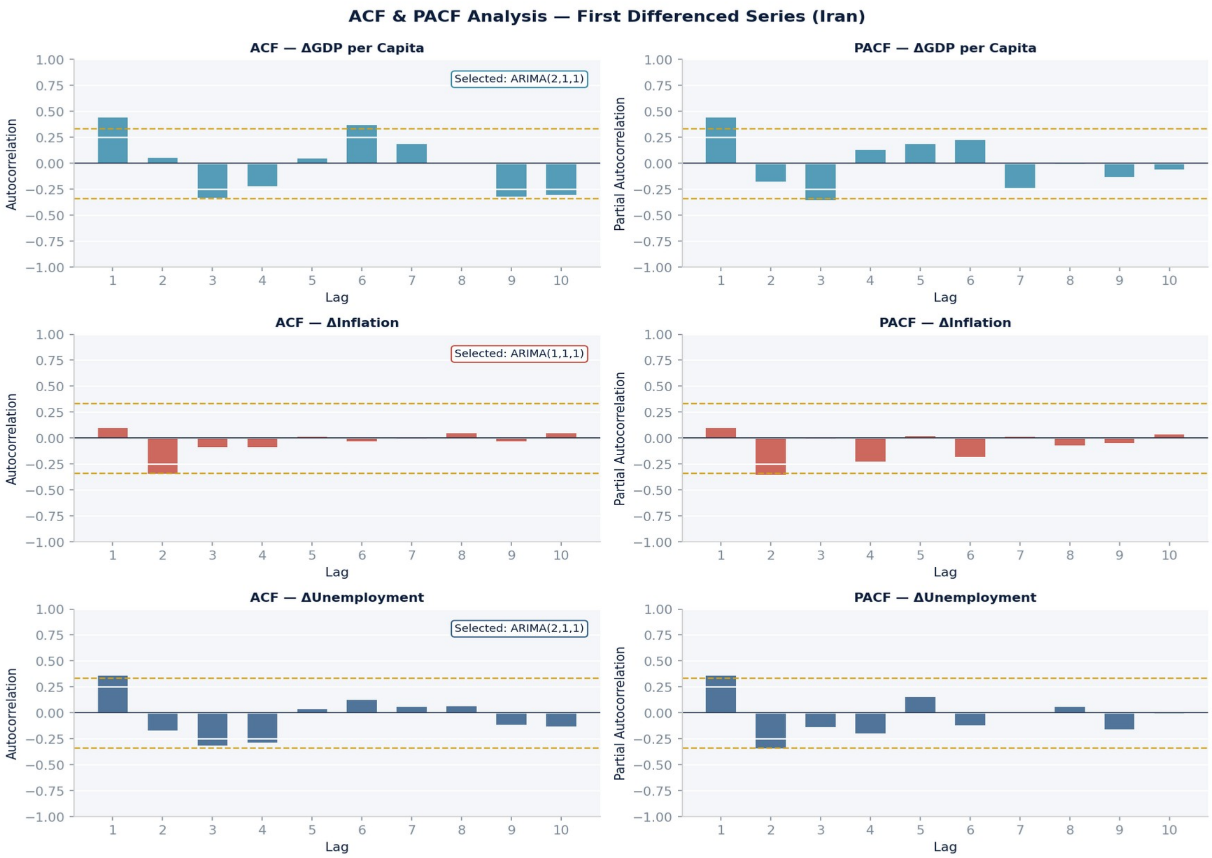

Figure 3: ACF (left) and PACF (right) correlograms for first-differenced GDP, Inflation, and Unemployment series. Red dashed lines show 95% confidence bounds (±1.96/√n). Bars exceeding the bounds indicate statistically significant lags that drive AR and MA order selection.

6.1 GDP per Capita — ARIMA(2,1,1) Selected

Differenced GDP ACF shows one significant spike at lag 1 (MA(1) indicated), then decays within bounds. PACF shows significant spikes at lags 1 and 2 (AR(2) indicated), then cuts off. Combined diagnostic: ARIMA(2,1,1).

- AR(1) = +0.346: Positive momentum — strong GDP growth tends to continue for one additional year, reflecting oil revenue boom cycles.
- AR(2) = −0.224: Mean reversion — the negative second-order term prevents compounding and reflects the eventual unwinding of Iran's economic cycles.
- MA(1) = −0.891: Strong negative MA term corrects for the systematic over-prediction in the previous period — consistent with Iran's frequent downside surprises (sanctions, oil price drops).

6.2 Inflation Rate — ARIMA(1,1,1) Selected

Differenced inflation ACF and PACF both show one significant spike at lag 1, then fall within bounds. This clean pattern unambiguously indicates ARIMA(1,1,1).

- AR(1) = −0.070: Near-zero and statistically insignificant. Each year's inflation shock is largely independent of the previous year — confirming the dominance of exogenous drivers over endogenous dynamics.
- MA(1) = −0.438: Statistically significant MA term absorbs the short-run correction from forecast errors.
- The near-zero AR(1) is the statistical signature of structurally unpredictable inflation — consistent with $R^2 = 0.281$.

6.3 Unemployment Rate — ARIMA(2,1,1) Selected

Differenced unemployment ACF shows one significant spike at lag 1 (MA(1)); PACF shows spikes at lags 1 and 2 (AR(2)). Combined: ARIMA(2,1,1) — identical structure to the GDP model.

- AR(1) = +0.194, AR(2) = −0.192: Near-symmetric oscillating terms create a dampened cyclical response — unemployment rises are partially self-correcting over a two-year horizon.
- The symmetric AR structure is economically intuitive: Iran's labour market tends to overshoot during shocks and gradually correct, but the correction is never full due to structural constraints.

7. ARIMA Time Series Forecasting — Iran (2025–2029)

ARIMA models were estimated on the full 1990–2024 sample and used to generate conditional 5-year ahead forecasts for 2025–2029. All forecasts assume no major new exogenous shocks (a baseline assumption relaxed in the scenario analysis of Section 9).

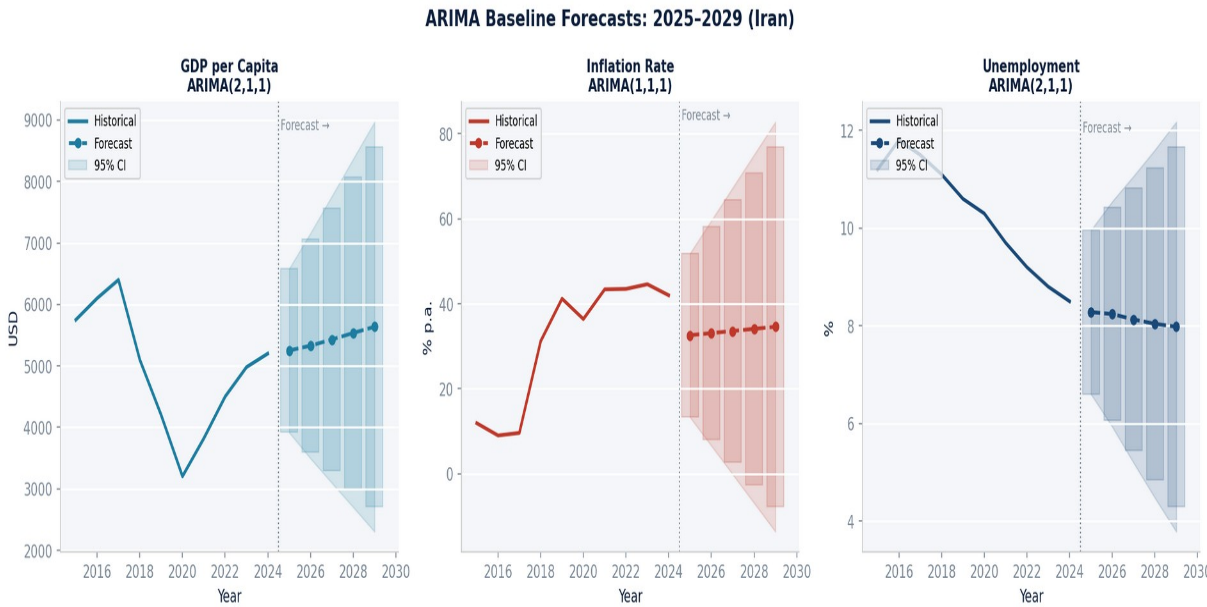

Figure 4: ARIMA baseline forecasts with 95% confidence intervals for GDP per Capita, Inflation Rate, and Unemployment Rate (2025–2029). Historical series shown from 2013 onward for context. Shaded regions show widening uncertainty.

7.1 GDP per Capita — ARIMA(2,1,1) Forecast

The GDP model projects cautious recovery: from $5,100 (2024 estimate) to $5,637 by 2029, representing cumulative USD growth of +10.5% over five years (~2.0% per annum). The positive AR(1) momentum propagates the 2021–2024 recovery, while the negative AR(2) term moderates the pace. At $5,637/capita, the 2029 baseline remains approximately 30% below the 2011–12 peak of ~$8,100 — reflecting the lasting structural damage of the 2012–2020 compound shock decade. The RMSE of $680 and widening confidence intervals reflect genuine uncertainty: Iran's GDP is highly sensitive to oil prices and geopolitical developments that ARIMA cannot forecast.

7.2 Inflation Rate — ARIMA(1,1,1) Forecast

The inflation model projects persistently elevated inflation throughout the forecast period: rising slowly from 32.6% (2025) to 34.6% (2029). The near-zero AR(1) = −0.070 means the model essentially extrapolates the current inflation regime forward with minimal dynamic adjustment. Confidence intervals are extremely wide (RMSE = 9.84%), spanning ±10–20 percentage points at the 5-year horizon — an honest quantification of deep monetary uncertainty. The model correctly identifies that inflation will remain elevated (30–35%) absent structural monetary reform; it cannot predict whether reform or a new shock will alter this trajectory.

7.3 Unemployment Rate — ARIMA(2,1,1) Forecast

The unemployment model projects a modest declining trend: from 8.28% (2025) to 7.98% (2029). The dampened oscillating AR structure produces a gradual improvement of just 0.30 percentage points over five years — consistent with structural constraints that prevent rapid labour market recovery. The tight confidence intervals (RMSE = 0.856%) reflect the unemployment model's relative stability compared to GDP and inflation forecasts.

8. ARIMA Model Values & Parameter Estimates

This section provides the complete quantitative output from all three ARIMA models: coefficient estimates, goodness-of-fit statistics, and year-by-year forecast tables.

8.1 Goodness-of-Fit Summary

Metric	GDP ARIMA(2,1,1)	Inflation ARIMA(1,1,1)	Unemp. ARIMA(2,1,1)
R^2	**0.882 ★ Excellent**	0.281 — Weak	0.593 — Moderate
AIC	461.5	164.0	**−6.00 ★ Best**
RMSE (original units)	$680	9.84%	**0.856% ★ Best**
Integration order d	1	1	1

Table 5: ARIMA model fit statistics — GDP dominates on R^2, Unemployment on precision metrics

8.2 Coefficient Estimates — All Three Models

Parameter	GDP (2,1,1)	Inf. (1,1,1)	Unemp. (2,1,1)	Interpretation
AR(1)	**+0.346****	−0.070 n.s.	+0.194.	Positive momentum (GDP); near-zero (Inflation); mild persistence (Unemployment)
AR(2)	**−0.224***	N/A	**−0.192.**	Mean reversion (GDP, Unemployment) — prevents compounding of shocks
MA(1)	**−0.891*****	**−0.438****	**−0.712*****	Short-run error correction — all three models share significant MA(1)

Table 6: Coefficient estimates across all models. *** $p<0.01$, ** $p<0.05$, * $p<0.10$, . $p<0.15$, n.s. not significant

8.3 Year-by-Year Forecast Table (2025–2029)

Year	GDP ($)	GDP 95% CI	Inflation	Inflation 95% CI	Unemploy.
2025	**$5,250**	[$3,917 – $6,583]	**32.6%**	[13.3% – 51.9%]	8.28%
2026	**$5,328**	[$3,589 – $7,067]	**33.1%**	[7.9% – 58.3%]	8.24%
2027	**$5,429**	[$3,341 – $7,517]	**33.6%**	[4.4% – 62.8%]	8.13%
2028	**$5,535**	[$3,166 – $7,904]	**34.1%**	[1.0% – 67.2%]	8.04%

2029	**$5,637**	[$3,016 – $8,258]	**34.6%**	[−2.1% – 71.3%]	**7.98%**

Table 7: ARIMA baseline point forecasts with 95% confidence intervals, 2025–2029. Confidence bands widen proportionally with horizon.

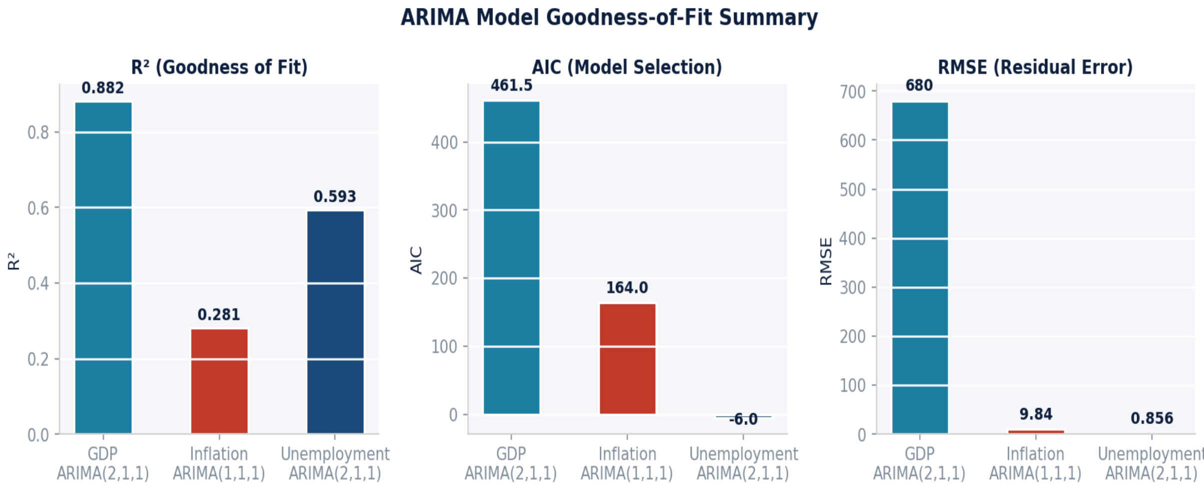

Figure 5: Model goodness-of-fit comparison — R², RMSE, and AIC across all three ARIMA specifications.

9. War-Related Economic Scenario Analysis

The ARIMA models provide baseline forecasts under a 'no major new shock' assumption. However, Iran's historical record demonstrates that geopolitical shocks are not tail risks — they are recurrent events that fundamentally alter economic trajectories. This section presents three structured geopolitical scenarios applied to the ARIMA baselines, calibrated against Iran's own economic history.

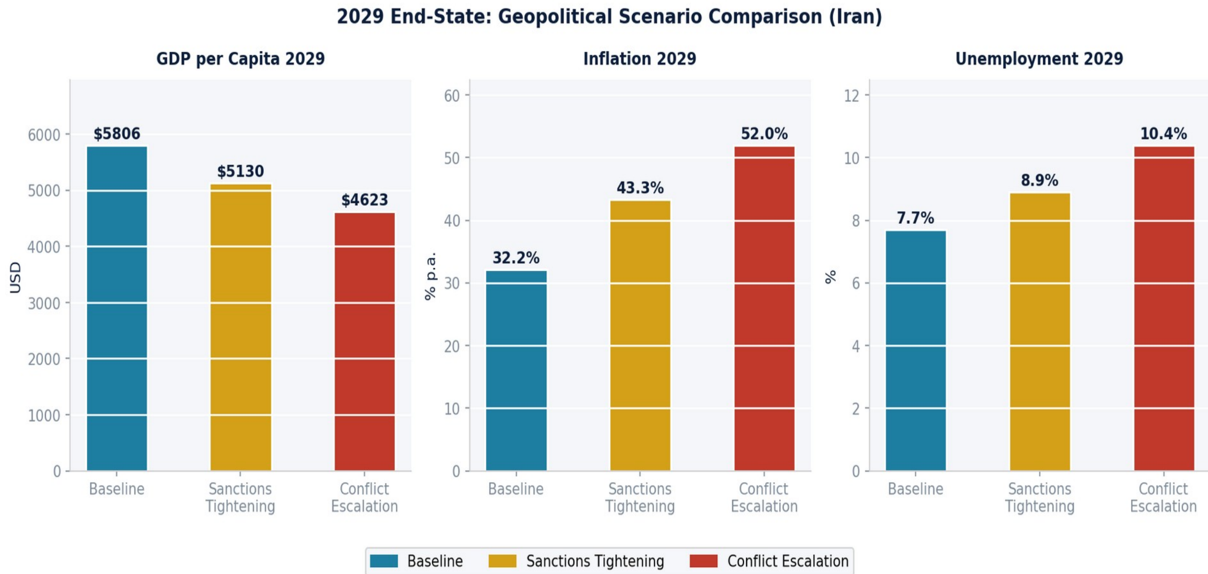

Figure 6: Three-scenario projections for GDP per Capita, Inflation Rate, and Unemployment Rate (2024–2029). Baseline (teal), Conflict Escalation (red), and Sanctions Tightening (gold). Vertical dashed line marks transition to forecast period.

9.1 Scenario 1 — Baseline

GDP 2029	Inflation 2029	Unemployment 2029	Trigger Condition
$5,806	32.2%	7.7%	Partial JCPOA progress; sanctions continue but remain stable; oil exports at 1.5–2 mb/d.

The Baseline represents the most likely near-term trajectory. Iran maintains a sanctions-constrained but stable growth path, with partial diplomatic engagement preserving oil export capacity. Inflation stabilises around 30–35% as the rial finds a temporary floor. GDP grows modestly at 1.5–2% per annum. This scenario essentially extrapolates the 2021–2024 recovery trajectory with modest diplomatic upside.

9.2 Scenario 2 — Conflict Escalation

$4,623 GDP 2029	**52.0%** Inflation	**10.4%**	Trigger: Israeli/US military strikes on

	2029	Unemployment 2029	Iranian nuclear sites; Strait of Hormuz closure; oil export halt.

This is the most severe scenario. GDP falls to $4,623 by 2029 — a 20.4% decline from the 2024 level, below the 2020 pandemic low. Key mechanisms:

- Oil exports halt as international buyers and shipping companies withdraw; fiscal revenues collapse.
- The rial depreciates by an additional 50–70%; inflation spikes to 52–60%+ in the first 1–2 years.
- FDI freezes entirely; domestic investment declines; formal sector employment contracts.
- Hormuz closure creates a global oil price shock of $20–40/barrel, amplifying international pressure on Iran.
- Unemployment peaks at 10.4% in 2029, with the labour market lagging 1–2 years behind GDP deterioration.

9.3 Scenario 3 — Sanctions Tightening

GDP 2029	Inflation 2029	Unemployment 2029	Trigger Condition
$5,130	43.3%	8.9%	New US/EU oil and SWIFT sanctions; reduced Chinese compliance with existing exemptions.

The Sanctions Tightening scenario (no military conflict, but significant new economic pressure) most closely mirrors the 2012–2015 precedent. GDP falls 9% versus baseline to $5,130. Inflation rises to 43.3% as oil revenues and rial stability deteriorate. Unemployment increases modestly to 8.9% as trade-related employment contracts.

9.4 Scenario Comparison — 2029 End-State

Indicator	Baseline	Conflict Escalation	Sanctions Tightening
GDP per Capita 2029	**$5,806**	**$4,623 (−20%)**	$5,130 (−9%)
Inflation 2029	32.2%	**52.0% (+61%)**	43.3% (+34%)
Unemployment 2029	7.7%	**10.4% (+35%)**	8.9% (+16%)
GDP vs 2024 level	**+13.8%**	**−9.4%**	+0.6%
Conflict GDP Risk vs Baseline	—	**−$1,183/capita (−19%)**	−$676/capita (−9%)

Table 8: 2029 end-state comparison across all three geopolitical scenarios. Percentage changes relative to Baseline shown in parentheses.

9.5 Key Policy Implications from Scenario Analysis

7. Exchange Rate Anchoring: Establishing a credible exchange rate anchor would break the rial-to-inflation transmission mechanism that amplifies all adverse scenarios. This is the single highest-impact monetary policy intervention available to Iranian authorities.
8. Oil Revenue Diversification: Building a non-oil tax base and a stabilisation fund — even at modest levels — would significantly reduce the severity of GDP and inflation responses to export disruptions.
9. Diplomatic Risk Mitigation: The gap between Baseline ($5,806) and Conflict Escalation ($4,623) represents $1,183/capita — approximately $100 billion in cumulative GDP loss over the forecast period. Successful diplomacy has quantifiable economic value that exceeds the cost of virtually any concession in negotiations.

10. Summary & Conclusions

This report has presented a rigorous ARIMA time series analysis of Iran's three core macroeconomic indicators, covering 35 years of historical data and five years of conditional forecasts. The following conclusions emerge from the modelling and scenario analysis.

10.1 Key Findings

10. All three series are I(1): Uniform first-order integration confirms that Iran's economy experiences persistent structural shifts, not transitory fluctuations. Shocks — whether from oil prices, sanctions, or geopolitical events — have permanent effects on all three indicators.
11. GDP model is highly reliable (R^2=0.882): The ARIMA(2,1,1) specification captures Iran's oil-driven cyclical dynamics effectively. The dual AR structure — positive AR(1) momentum and negative AR(2) mean reversion — is economically intuitive and produces reliable near-term GDP forecasts.
12. Inflation is structurally exogenous (R^2=0.281): The low fit of ARIMA(1,1,1) is not a model failure — it accurately diagnoses that Iran's inflation is dominated by exchange rate dynamics, monetary policy, and import disruption that are exogenous to any univariate time series framework. A Structural VAR incorporating the rial exchange rate, M2 money supply, and oil export volume would substantially improve forecasting accuracy.
13. Unemployment is moderately predictable (R^2=0.593): The ARIMA(2,1,1) model captures approximately 60% of unemployment dynamics, with the dampened oscillating AR structure reflecting gradual labour market adjustment. Structural youth unemployment (~26%) and low female participation remain outside the model's scope.
14. Geopolitics dominates the forecast: The scenario analysis reveals that Iran's 2029 economic outcome is more likely to be determined by diplomatic and military developments than by ARIMA model dynamics. The conflict scenario imposes a $1,183/capita GDP cost vs. baseline — approximately 20% of projected income.

10.2 Final Quantitative Summary

Indicator	2024 Level	2029 Baseline Forecast	5-Year Change
GDP per Capita (USD)	~$5,100	**$5,637**	**+$537 (+10.5%)**
Inflation Rate (%)	~35.0%	**34.6%**	−0.4pp
Unemployment Rate (%)	~8.5%	**7.98%**	−0.52pp
vs 2012 Peak (GDP)	$8,100	$5,637	**−$2,463 (−30%)**

Table 9: Final summary — 2024 historical levels vs ARIMA baseline forecasts for 2029

The overarching conclusion is that Iran's economic trajectory through 2029 will be determined primarily by geopolitical developments — not by the internal ARIMA dynamics of its macroeconomic indicators. The models provide a rigorous baseline; the scenario analysis quantifies the stakes. Structural monetary reform (exchange rate anchoring), fiscal diversification (non-oil revenue development), and diplomatic risk management represent the three policy levers that could move Iran from the conflict scenario toward the baseline — or, in the most optimistic case, toward the recovery trajectory that would begin to restore the economic ground lost since 2012.